



**COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF SCIENCE
DEPARTMENT OF MATHEMATICS
APPLIED MATHEMATICS**

MODULE: Final Year Project

**Application of Frobenius Method for solving ordinary
Differential Equation Involving Variable Coefficients**

This Final Year Project is submitted to the College of Science and Technology in partial fulfillment of the requirements for the award of a Bachelor's Degree with Honours in Applied Mathematics.

Name: USENGIMANA Richard
Reg No: 222010997
Supervisor: Dr. Venuste NYAGAHAKWA
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Dedication

I sincerely declare that I am the sole writer of this project and it is personally prepared by me to be submitted in Partial Fulfillment of the Academic Degree of Bachelor of Science in Applied Mathematics

Names: USENGIMANA Richard

Reg no: 222010997

Signature:.....

Certification

I certify that this final year project is my work under the organizational supervision and is the original copy. .

Supervisor:**Dr. Venuste NYAGAHAKWA**

Signature:

.

Head Of Department: **Dr. Celestin KURUGIBWAMI**

Signature:

Submission Date:

Acknowledgment

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Abstract

This research is based on the Frobenius method as one of the analytical techniques for solving ordinary differential equations (ODEs) with variable coefficients. We will explore the structure and theory behind the Frobenius method, which is designed to find series solutions near regular singular points of second-order linear differential equations. We will go through the formulation of the Frobenius series, derive the indicial equation, allowing solutions even when the differential equation has a singularity at the point of expansion. Due to the structure of the Frobenius series we will apply the Frobenius method to Legendre's differential equation. This example will demonstrate how the Frobenius approach can be used to obtain polynomial solutions (Legendre polynomials) and highlight the method's practical applications in solving real-world problems.

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Chapter 1

GENERAL INTRODUCTION

1.1 Introduction

Differential equations play a fundamental role in describing a wide range of natural and engineered systems. Mathematically, a **differential equation (DE)** is an equation that involves one or more functions and their derivatives, representing the relationship between a quantity and its rate of change. These equations are essential in modeling physical phenomena such as heat conduction, wave propagation, fluid dynamics, population growth, and electrical circuits[1][2].

A differential equation is classified as an **ordinary differential equation (ODE)** if it involves derivatives with respect to a single independent variable. A general form of an n th-order linear ODE is given by:

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \quad (1)$$

where $a_i(x)$ for $i = 0, 1, \dots, n$ are functions of the independent variable x , and $g(x)$ is the nonhomogeneous part (which may be zero in the case of homogeneous equations)[1]. When the coefficients $a_i(x)$ are constants, many standard solution techniques such as the method of undetermined coefficients or variation of parameters can be employed effectively. However, when these coefficients are variable functions, especially when they are not analytic at certain points, solving the equation becomes more complex.

In such cases, **power series methods** provide a powerful tool for finding solutions in the neighborhood of a point. However, standard power series solutions

are only valid around **ordinary points**, where all coefficients are analytic. When the point of interest is a **singular point**, particularly a **regular singular point**, the usual power series method breaks down.

To address this, the **Frobenius method** is employed. It is an extension of the power series method that allows for solutions in the form:

$$y(x) = (x - x_0)^r \sum_{k=0}^{\infty} a_k (x - x_0)^k, \quad (2)$$

where r is a number to be determined (not necessarily an integer), and x_0 is the regular singular point[2][3]. The method is particularly well-suited for solving second-order or higher-order linear ODEs where the coefficients are not analytic but satisfy certain conditions around the singular point. Specifically, if x_0 is a regular singular point, which means that the functions

$$(x - x_0) \cdot \frac{a_{n-1}(x)}{a_n(x)}, \quad (x - x_0)^2 \cdot \frac{a_{n-2}(x)}{a_n(x)}, \quad \dots, \quad (x - x_0)^n \cdot \frac{a_0(x)}{a_n(x)} \quad (3)$$

are analytic at x_0 , then the Frobenius method becomes applicable.

1.2 Problem statement

Many physical and engineering systems are governed by ODEs with variable coefficients, arising in disciplines such as quantum mechanics, fluid dynamics, electromagnetism, and structural analysis[1]. This project explores the application of the Frobenius method in solving differential equations with variable coefficients, emphasizing both its theoretical foundation and practical significance. Many physical and engineering systems such as those found in quantum mechanics, fluid dynamics, electromagnetism, and structural analysis are governed by ODEs with variable coefficients. Unlike constant coefficient ODEs, these equations often defy traditional solution techniques like separation of variables, integrating factors, or simple power series expansions, especially near singular points.

To address these challenges, this study applies the Frobenius method to notable examples from mathematical physics. The method proves especially valuable in contexts where conventional approaches fail, offering accurate and analytical solutions near singularities. Through real-world applications drawn from engineering systems and physical processes, the study demonstrates how the Frobenius method is not only applicable but often essential for solving variable-coefficient ODEs.

- **Legendre's equation:**

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0,$$

which appears in spherical coordinate systems, particularly in potential theory and solutions to Laplace's equation.

These equations possess (1) **regular singular points** at $x = 0$ (for Bessel) and $x = \pm 1$ (for Legendre), making the Frobenius method highly suitable for deriving solutions in the neighborhood of those points. By solving these canonical equations, the project aims to illustrate the power, scope, and limitations of the Frobenius method in handling ODEs with variable coefficients that commonly occur in physical and engineering contexts.

1.3 Objectives of the study

1.3.1 Main objective

The main Objective is To analyze and apply the Frobenius method for solving ODEs with variable coefficients, demonstrating its effectiveness in real-world applications in physics and engineering.

1.3.2 Specific objectives

the specific Objectives of this project is

- To implement the Frobenius method for solving the Legendre equation
- To compute the solutions obtained by the Frobenius method by using computational tools such as Python.
- To compare the analytical solutions obtained using the Frobenius method with the numerical solutions computed using Python.

1.4 Methodology

This project employs a structured analytical and computational approach to examine and apply the Frobenius method for solving ODEs with variable coefficients. The methodology consists of the following key steps:

1. **Analytical Implementation of the Frobenius Method:** In this project, the Frobenius method will be systematically applied to the Legendre differential equation and identifying the regular singular point of the equation, which is essential .
- 2.

1.5 Significance of the Study

The study of the Frobenius method is crucial for solving ordinary differential equations with variable coefficients, particularly those exhibiting singularities, which arise in various scientific and engineering applications. By systematically analyzing and applying this method, the research improves understanding of power series solutions and their convergence properties. The findings will provide valuable information for mathematicians, physicists, and engineers working on problems in quantum mechanics, fluid dynamics, electromagnetism, and structural analysis.

1.6 Organization of Work

This project is organized into five main chapters. The first chapter provides a general introduction to the topic, including the background of the study, the problem statement, objectives, significance, and the structure of the work. The second chapter presents the theoretical background of ODEs. It covers the basic definitions, classifications, and common methods used to solve ODEs, with a focus on the challenges posed by equations with variable coefficients. The third chapter discusses the power series, a powerful technique for solving second-order linear ODEs with variable coefficients near regular singular points. This chapter explains the method in detail, supported by work-in-progress examples to demonstrate its application. The fourth chapter deal with the application of Frobenious method for solving Legendre equation and the last chapeter that contain the conclusion and recommendations

Chapter 2

Theory of Ordinary differential equation

2.1 Ordinary differential equations

The Term **Differential equation** was first used by **Leibniz** in **1676** to denote a relationship between the differential dx and dy of two variables x and y . In general, this relationship explicitly involves the variables x and y together with other symbols $a, b, c, \dots$ that represent constants.

Differential equations are classified in the first place, according to the number of variables which they involve. An ordinary differential equation expresses a relation between an independent variable, a dependent and or more differential coefficients of the dependent with respect to the independent variable[8]. The function typically depends on a single independent variable, often denoted as x or t .

In general, an ODE is written as:

$$F(x, y, y', y'', \dots, y^{(n)}) = 0,$$

where $y^{(n)}$ is the n -th derivative of the unknown function y , and F is some function involving x, y , and its derivatives.

2.2 Classification of ordinary differential Equations

Ordinary differential equations are primarily classified according to their **order** and **linearity**. The **order** of a differential equation refers to the highest derivative of the dependent variable present in the equation. For instance, the equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ is a second-order differential equation because the highest derivative involved is of order two.

An n th-order ordinary differential equation is said to be **linear** if it is linear in the dependent variable y and all of its derivatives up to the n th order. In other words, the equation must take the form:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x),$$

where the coefficients $a_0(x), a_1(x), \dots, a_n(x)$ are functions of the independent variable x , and $g(x)$ is a given function. For lower-order cases, such as first and second-order differential equations, the corresponding forms are:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \quad a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x).$$

In all these cases, the dependent variable y and its derivatives $y', y'', \dots, y^{(n)}$ appear only to the first power, which means that each term involving y or its derivatives is of degree one. Furthermore, the coefficients of these terms are functions of the independent variable x alone and do not depend on y or its derivatives.

A homogeneous linear differential equation of order n is an equation of the form:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

In other words, “homogeneous” just means that $g(x) = 0$. Otherwise it is said to be nonhomogeneous.

2.3 Different method used for Solving Ordinary Differential Equations

There are various methods for solving ODEs depending on the type and order of the equation. Below are common methods:

2.3.1 Separation of Variables (First-order ODEs)

We now examine a solution technique for finding exact solutions to a class of differential equations known as separable differential equations. These equations are common in a wide variety of disciplines, including physics, chemistry, and engineering.

This method works when the ODE can be written in the form:

$$\frac{dy}{dx} = g(x)h(y)$$

where the derivative $\frac{dy}{dx}$ is a product of a function of x alone times a function of y alone, rather than a more general function of the two variables x and y .

Remark 2.3.1 *A separable differential equation is $y' = h(y)g(x)$ has the following properties:*

- *The left-hand side depends explicitly only on y , so any x dependence is through y .*
- *The right-hand side depends only on x .*
- *And the left-hand side is of the form (something on y) $\times y'$*

.

Examples 2.3.1 *Find all solutions y to the differential equation*

$$-\frac{y'}{y^2} = \cos(2t).$$

Solution 2.3.1 *The differential equation above is separable, with*

$$g(t) = \cos(2t), \quad h(y) = -\frac{1}{y^2}.$$

Therefore, it can be integrated as follows:

$$-\frac{y'}{y^2} = \cos(2t) \quad \Leftrightarrow \quad \int \frac{-y'(t)}{y^2(t)} dt = \int \cos(2t) dt + c.$$

The integral on the right-hand side can be computed explicitly. The integral on the left-hand side can be done by substitution. The substitution is

$$u = y(t), \quad du = y'(t) dt \Rightarrow \int \frac{-du}{u^2} dt = \int \cos(2t) dt + c.$$

This notation makes clear that u is the new integration variable, while $y(t)$ are the unknown function values we look for. However, it is common in the literature to use the same name for the variable and the unknown function. We will follow that convention, and we write the substitution as

$$y = y(t), \quad dy = y'(t) dt \Rightarrow \int \frac{-dy}{y^2} = \int \cos(2t) dt + c.$$

We hope this is not too confusing. Integrating on both sides above we get

$$\frac{1}{y} = \frac{1}{2} \sin(2t) + c.$$

So, we get the implicit and explicit form of the solution,

$$\frac{1}{y(t)} = \frac{1}{2} \sin(2t) + c \Leftrightarrow \boxed{y(t) = \frac{2}{\sin(2t) + 2c}}.$$

2.3.2 Exact ordinary differential equations

We recall from calculus that if a function $u(x, y)$ has continuous partial derivatives, its differential (also called its total differential) is

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

From this it follows that if $u(x, y) = c$ where c is a constant, then $du = 0$

Example of first-order ODE $M(x, y) + N(x, y)y' = 0$ written as (use $dy = y'dx$)

$$M(x, y)dx + N(x, y)dy = 0$$

is called an exact differential equation if the differential form $M(x, y)dx + N(x, y)dy$ is exact, that is, this form is the differential

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

of some function $u(x, y)$

Example (Exact ODE): Solve the following Ordinary Differential Equation:

$$\cos(x + y)dx + (3y^2 + 2y + \cos(x + y))dy = 0 \quad (2.1)$$

(i) Test for exactness: Our equation is of the form (1.4.1) with $M = \cos(x + y)$ and $N = 3y^2 + 2y + \cos(x + y)$. It follows that

$$\frac{\partial M}{\partial y} = -\sin(x + y), \quad \frac{\partial N}{\partial x} = -\sin(x + y).$$

From this and (1.4.5) we see that (1.4.8) is exact.

(ii) Implicit general solution: From (1.4.6) we obtain by integration

$$u = \int Mdx + k(y) = \int \cos(x + y)dx + k(y) = \sin(x + y) + k(y) \quad (1.4.9)$$

To find $k(y)$, we differentiate this formula with respect to y and use formula (1.4.4)(b), obtaining

$$\frac{\partial u}{\partial y} = \cos(x + y) + \frac{dk}{dy} = N = 3y^2 + 2y + \cos(x + y)$$

Hence

$$\frac{dk}{dy} = 3y^2 + 2y.$$

By integrating, we get $k(y) = y^3 + y^2 + c_1$. Inserting this result into (1.4.9) and observing (1.4.3), we obtain the answer

$$u(x, y) = \sin(x + y) + y^3 + y^2 = c$$

(iii) Checking an implicit solution: We can check by differentiating the implicit solution $u(x, y) = c$ implicitly and see whether this leads to the given ODE (1.4.8):

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = \cos(x + y)dx + [\cos(x + y) + 3y^2 + 2y]dy = 0 \quad (1.4.10)$$

2.3.3 Bernoulli's Equation

In 1696 Jacob Bernoulli solved what is now known as the Bernoulli differential equation. This is a first order nonlinear differential equation[1]. The following year Leibniz solved this equation by transforming it into a linear equation.

Definition 2.3.1 *the **bernoulli equation** [7] is*

$$y' = P(t)y + q(t)y^n$$

where p, q are given functions and $n \in \mathbb{R}$.

Remark 2.3.2

- (a) For $n \neq 0, 1$, the equation is nonlinear.
- (b) If $n = 2$ we get the logistic equation, (we'll study it in a later chapter),

$$y' = ry(1 - \frac{y}{K})$$

(c) This is not the Bernoulli equation from fluid dynamics. The Bernoulli equation is special in the following sense: it is a nonlinear equation that can be transformed into a linear equation.

Theorem 2.3.1 (Bernoulli). *The function y is a solution of the Bernoulli equation:*

$$y' = p(t)y + q(t)y^n, \quad n \neq 1,$$

if and only if the function $v = \frac{1}{y^{n-1}}$ is a solution of the linear differential equation:

$$v' = -(n-1)p(t)v - (n-1)q(t).$$

Remark 2.3.3 *This result summarizes Laplace's idea to solve the Bernoulli equation. It transforms the nonlinear Bernoulli equation for y into a linear equation for $v = \frac{1}{y^{n-1}}$. One then solves the linear equation for v using the integrating factor method. The final step is to transform back to $y = \left(\frac{1}{v}\right)^{\frac{1}{n-1}}$.*

Proof : Divide the Bernoulli equation by y^n :

$$\frac{y'}{y^n} = \frac{p(t)}{y^{n-1}} + q(t).$$

Introduce the new variable $v = y^{-(n-1)}$, and compute its derivative:

$$v' = (y^{-(n-1)})' = -(n-1)y^{-n}y' \Rightarrow -\frac{v'(t)}{n-1} = \frac{y'(t)}{y^n(t)}.$$

If we Substitute v and this last equation into the Bernoulli equation we get:

$$-\frac{v'}{n-1} = p(t)v + q(t) \Rightarrow v' = -(n-1)p(t)v - (n-1)q(t).$$

This completes the proof.

Examples 2.3.2 Find every solution of the equation $ty' = 3y + t^5y^{\frac{1}{3}}$

Solution 2.3.2 Rewrite the differential equation as

$$y' = \frac{3}{t}y + t^4y^{\frac{1}{3}}$$

This is a Bernoulli equation for $n = \frac{1}{3}$. Divide the equation by $y^{\frac{1}{3}}$ we get ,

$$\frac{y'}{y^{\frac{1}{3}}} = \frac{3}{t}y^{\frac{2}{3}} + t^4$$

Now, Define the new unknown function $v = \frac{1}{y^{\frac{1}{3}}}$, that is $v = y^{\frac{2}{3}}$, compute the derivative $v' = \frac{2y'}{3y^{\frac{1}{3}}}$ and introduce them in the differential equation

$$\frac{3}{2}v' = \frac{3}{t}v + t^4 \Rightarrow v' - \frac{2}{t}v = \frac{2}{3}t^4$$

this is a linear equation for v . Integrate this equation using the integrating factor method. To compute the integrating factor we need to find

$$A(t) = \int \frac{2}{t} dt = 2\ln(t) = \ln(t^2)$$

Then the integrating factor is $\mu(t) = e^{-A(t)}$. In this case we get

$$\mu(t) = e^{\ln(t^2)} = e^{\ln(t^{-2})} \Rightarrow \mu(t) = \frac{1}{t^2}$$

Therefore, the equation for v can be written as a total derivative,

$$\frac{1}{t^2}(v' - \frac{2}{t}v) = \frac{2}{3}t^2 \Rightarrow (\frac{v}{t^2} - \frac{2}{9}t^3)' = 0$$

The potential function is $\psi(t, v) = \frac{v}{t^2} - (2/9)t^3$ and the solution of the differential equation is $\psi(t, v(t)) = c$, that is

$$\frac{v}{t^2} - \frac{2}{9}t^3 = c \Rightarrow v(t) = t^2(c + \frac{2}{9}t^3) \Rightarrow v(t) = ct^2 + \frac{2}{9}t^5$$

Once v is known we compute the original unknown $y = \pm v^{\frac{3}{2}}$, where the double sign is related to taking the square root.

We finally obtain the following.

$$y(t) = \pm(ct^2 + \frac{2}{9}t^5)^{\frac{3}{2}}$$

2.3.4 Second Order Linear Equations

In the previous section we show a few types of ODEs that can be solved in terms of familiar elementary functions. The methods we developed require considerable skill in the integration techniques, and their many interesting applications have a tasty flavor of practicality. Unfortunately, however, it must be admitted that this part of the subject tends to be a miscellaneous bag of tricks and conveys little insight into the general nature of differential equations and their solutions[3]. We shall see that this theory can be given a coherent and satisfying structure based on a few simple principles.

The general second-order linear differential equation is

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x), \quad (17)$$

or more simply,

$$y'' + P(x)y' + Q(x)y = R(x) \quad (18)$$

As the notation indicates, it is understood that $P(x)$, $Q(x)$, and $R(x)$ are functions of x alone (or perhaps constants). It is clear that no loss of generality results

from taking the coefficient of y'' to be 1, since this can always be accomplished by division[3]. Equations of this kind are of great significance in physics, especially in connection with vibrations in mechanics and the theory of electric circuits[9]. In addition, many profound and beautiful ideas in pure mathematics have grown out of the study of these equations.

Theorem 2.3.2 *If $y_1(x)$ and $y_2(x)$ are any two solutions of (18), then*

$$c_1 y_1(x) + c_2 y_2(x)$$

is also a solution for any constants c_1 and c_2 .

The statement follows immediately from the fact that

$$\begin{aligned} & (c_1 y_1 + c_2 y_2)'' + P(x)(c_1 y_1 + c_2 y_2)' + Q(x)(c_1 y_1 + c_2 y_2) \\ &= (c_1 y_1'' + c_2 y_2'') + P(x)(c_1 y_1' + c_2 y_2') + Q(x)(c_1 y_1 + c_2 y_2) \\ &= c_1 [y_1'' + P(x)y_1' + Q(x)y_1] + c_2 [y_2'' + P(x)y_2' + Q(x)y_2] \\ &= c_1 \cdot 0 + c_2 \cdot 0 = 0, \end{aligned}$$

where the multipliers of c_1 and c_2 are zero because, by assumption, y_1 and y_2 are solutions of (18).

2.3.5 Constant Coefficients

We studied first-order linear equations, where we obtained a formula for all solutions to these equations. We could say that we know all that can be known about solutions to first-order linear equations. However, this is not the case for solutions to second-order linear equations since we do not have a general formula for all solutions to these equations.

In this section we present two main results, which says that there are solutions to second order linear equations when the equation coefficients are continuous functions. Furthermore, these solutions have two free parameters that can be fixed by appropriate initial conditions.

Definition 2.3.2 *A second order linear differential equation for the function y is[7]*

$$y'' + a_1(t)y' + a_0(t)y = b(t), \quad (19)$$

where a_1, a_0, b are given functions on the interval $I \subset \mathbb{R}$. The Eq. (19) above:

- is **homogeneous** iff the source $b(t) = 0$ for all $t \in \mathbb{R}$;
- has **constant coefficients** iff a_1 and a_0 are constants;
- has **variable coefficients** iff either a_1 or a_0 is not constant.

Example: Find the differential equation satisfied by the family of functions

$$y(t) = c_1 e^{4t} + c_2 e^{-4t},$$

where c_1, c_2 are arbitrary constants.

Solution: From the definition of y compute c_1 ,

$$c_1 = y e^{-4t} - c_2 e^{-8t}.$$

Now compute the derivative of function y

$$y' = 4c_1 e^{4t} - 4c_2 e^{-4t},$$

Replace c_1 from the first equation above into the expression for y' ,

$$y' = 4(y e^{-4t} - c_2 e^{-8t}) e^{4t} - 4c_2 e^{-4t} \Rightarrow y' = 4y + (-4 - 4)c_2 e^{-4t},$$

so we get an expression for c_2 in terms of y and y' ,

$$y' = 4y - 8c_2 e^{-4t} \Rightarrow c_2 = \frac{1}{8}(4y - y') e^{4t}$$

At this point we can compute c_1 in terms of y and y' , although we do not need it for what follows. Anyway,

$$c_1 = y e^{-4t} - \frac{1}{8}(4y - y') e^{4t} e^{-8t} \Rightarrow c_1 = \frac{1}{8}(4y + y') e^{-4t}.$$

We do not need c_1 because we can get a differential equation for y from the equation for c_2 .

Compute the derivative of that equation,

$$0 = c_2' = \frac{1}{8}(4y - y') e^{4t} + \frac{1}{8}(4y' - y'') e^{4t} \Rightarrow 4(4y - y') + (4y' - y'') = 0$$

which gives us the following second order linear differential equation for y ,

$$\boxed{y'' - 16y = 0}.$$

Here's example for Solve the non-homogeneous differential equation:

$$y'' - 3y' + 2y = e^x$$

Solve the homogeneous equation

$$y'' - 3y' + 2y = 0$$

The characteristic equation is:

$$r^2 - 3r + 2 = 0 \Rightarrow r = 1, \quad r = 2$$

So the general solution to the homogeneous equation is:

$$y_h(x) = C_1 e^x + C_2 e^{2x}$$

Find a particular solution $y_p(x)$

Guess:

$$y_p(x) = A x e^x$$

Compute derivatives:

$$y_p' = A(1+x)e^x, \quad y_p'' = A(2+x)e^x$$

Substitute into the original equation:

$$y_p'' - 3y_p' + 2y_p = e^x$$

$$A(2+x)e^x - 3A(1+x)e^x + 2Axe^x = e^x$$

Factor Ae^x :

$$Ae^x[(2+x) - 3(1+x) + 2x] = e^x$$

Simplify:

$$Ae^x(-1) = e^x \Rightarrow A = -1$$

Particular solution:

$$y_p(x) = -xe^x$$

General solution

$$y(x) = y_h(x) + y_p(x) = C_1 e^x + C_2 e^{2x} - xe^x$$

$$\boxed{y(x) = C_1 e^x + C_2 e^{2x} - xe^x}$$

2.3.6 Solutions to the Initial Value Problem

Here is the first of the two main results in this section. Second order linear differential equations have solutions in the case that the equation coefficients are continuous functions. Since the solution is unique when we specify two initial conditions, the general solution must have two arbitrary integration constants.

Theorem 2.3.3 (IVP) *the functions a_1, a_0, b are continuous on a closed interval $I \subset \mathbb{R}$, the constant $t_0 \in I$, and $y_0, y_1 \in \mathbb{R}$ are arbitrary constants, then there is a unique solution y , defined on I , of the initial value problem*

$$\begin{aligned}y'' + a_1(t)y' + a_0(t)y &= b(t), \\ y(t_0) &= y_0, \quad y'(t_0) = y_1.\end{aligned}$$

Examples 2.3.3 *Find the domain of the solution to the initial value problem*

$$(t-1)y'' - 3ty' + \frac{4(t-1)}{t-3}y = t(t-1), \quad y(2) = 1, \quad y'(2) = 0.$$

Solution: We first write the equation above in the form given in the Theorem above:

$$y'' - \frac{3t}{t-1}y' + \frac{4}{t-3}y = t.$$

The equation coefficients are defined on the domain

$$(-\infty, 1) \cup (1, 3) \cup (3, \infty).$$

So the solution may not be defined at $t = 1$ or $t = 3$. That is, the solution is defined in

$$(-\infty, 1), \quad (1, 3), \quad \text{or} \quad (3, \infty).$$

Since the initial condition is at $t_0 = 2 \in (1, 3)$, then the domain of the solution is

$$D = (1, 3).$$

2.3.7 Differential Equation Involving Valuable Coefficients

Differential equations with variable coefficients arise frequently in physics, engineering, and economics. These equations differ from their constant coefficient counterparts in that the functions multiplying the dependent variable and its derivatives are not constant, but rather depend on the independent variable. A classical form of such an equation is the second-order linear differential equation:

$$y'' + a_1(t)y' + a_0(t)y = b(t),$$

where $a_1(t)$, $a_0(t)$, and $b(t)$ are continuous functions defined on some interval. If $b(t) = 0$, the equation is called homogeneous; otherwise, it is nonhomogeneous.

A key challenge in solving such equations lies in the lack of a general closed-form solution, unlike the constant coefficient case. Still, various theoretical tools such as the superposition principle, the Wronskian, and Abel's theorem provide powerful ways to understand the structure and behavior of solutions.

Existence and Uniqueness of Solutions

One foundational result is the existence and uniqueness theorem. It states that if $a_1(t)$, $a_0(t)$, and $b(t)$ are continuous on an interval containing t_0 , and initial conditions $y(t_0) = y_0$, $y'(t_0) = y_1$ are given, then a unique solution exists in that interval. This guarantees that under reasonable conditions, a well-posed problem will yield a unique trajectory for the system's behavior.

Example 1: Determining the Interval of Solution

Consider the equation:

$$(t - 1)y'' - 3ty' + 4y = t(t - 1), \quad y(-2) = 2, \quad y'(-2) = 1.$$

Rewriting it in standard form gives:

$$y'' - \frac{3t}{t-1}y' + \frac{4}{t-1}y = t.$$

The coefficients are discontinuous at $t = 1$, so the largest interval where the functions are continuous and which contains the initial point $t = -2$ is $(-\infty, 1)$. Thus, a unique solution exists in this interval.

2.3.8 Wronskian Methods

We now introduce a function that provides important information about the linear dependency of two functions y_1, y_2 . This function, W , is called the Wronskian to

honor the polish scientist Josef Wronski, who first introduced this function in 1821 while studying a different problem[?].

The Wronskian of the differentiable functions y_1 and y_2 is the function

$$W_{12}(t) = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Remark: Introducing the matrix-valued function

$$A(t) = \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix},$$

the Wronskian can be written using the determinant of that 2×2 matrix:

$$W_{12}(t) = \det(A(t)).$$

An alternative notation is:

$$W_{12}(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}.$$

Examples 2.3.4 Let $y_1(t) = \sqrt{t}$, $y_2(t) = \frac{1}{t}$. We test whether these are independent solutions of the equation:

$$2t^2y'' + 3ty' - y = 0.$$

First compute the Wronskian:

$$W(t) = \begin{vmatrix} \sqrt{t} & \frac{1}{t} \\ \frac{1}{2\sqrt{t}} & -\frac{1}{t^2} \end{vmatrix} = \sqrt{t} \cdot \left(-\frac{1}{t^2}\right) - \frac{1}{2\sqrt{t}} \cdot \frac{1}{t} = -\frac{1}{t^{3/2}} - \frac{1}{2t^{3/2}} = -\frac{3}{2t^{3/2}}.$$

Since $W(t) \neq 0$, the solutions are linearly independent and thus form a fundamental set.

Abel's Theorem and the Wronskian

Abel's theorem provides a powerful result: if y_1 and y_2 are solutions of the homogeneous second-order linear ODE, then their Wronskian satisfies the differential equation:

$$W'(t) + a_1(t)W(t) = 0.$$

This can be integrated using an integrating factor to yield:

$$W(t) = W(t_0)e^{-\int_{t_0}^t a_1(s) ds}.$$

This relation reveals that if the Wronskian is zero at any point, it is zero everywhere, implying linear dependence.

Special Second-Order Nonlinear Equations

Not all second-order equations are linear. Nonlinear second-order ODEs occur when the function $f(t, y, y')$ in the general form $y'' = f(t, y, y')$ is nonlinear in y or y' . These are often more challenging to solve. However, there are two important cases where nonlinear equations can be transformed into first-order equations.

Case (a): If the equation is of the form $y'' = f(t, y')$, the dependent variable y does not appear explicitly. Define $v = y'$, then $v' = f(t, v)$, a first-order ODE in v .

Case (b): If the equation is of the form $y'' = f(y, y')$, the independent variable t does not appear. If y is invertible, we define $v(t) = y'(t)$, and $\hat{v}(y) = v(t(y))$. Then:

$$\frac{d\hat{v}}{dy} = \frac{1}{\hat{v}} f(y, \hat{v}).$$

This transforms the second-order ODE into a first-order ODE in $\hat{v}(y)$.

Example 4: Missing y

Solve the nonlinear equation:

$$y'' = -2t(y')^2, \quad y(0) = 2, \quad y'(0) = 1.$$

Let $v = y'$, then $v' = -2tv^2$. This is separable:

$$\frac{dv}{v^2} = -2t dt \quad \Rightarrow \quad \int v^{-2} dv = \int -2t dt.$$

Integrating both sides:

$$-\frac{1}{v} = -t^2 + C \Rightarrow \frac{1}{v} = t^2 - C.$$

Now, using the initial condition $v(0) = 1 \Rightarrow \frac{1}{1} = -C \Rightarrow C = -1$. Then:

$$v = y' = \frac{1}{t^2 + 1}.$$

Integrate y' to find y :

$$y = \int \frac{1}{t^2 + 1} dt = \tan^{-1}(t) + D.$$

With $y(0) = 2 \Rightarrow D = 2$. So:

$$y(t) = \tan^{-1}(t) + 2.$$

Example 5: Missing t

Solve the nonlinear equation:

$$y'' = 2yy', \quad \text{no explicit } t.$$

Let $v = y' \Rightarrow v' = 2yv$. Define $\hat{v}(y) = v(t(y))$, then:

$$\frac{d\hat{v}}{dy} = \frac{1}{\hat{v}} \cdot 2y\hat{v} = 2y \Rightarrow \hat{v} = y^2 + C.$$

So:

$$v = y' = y^2 + C.$$

This is a separable equation:

$$\frac{dy}{y^2 + C} = dt.$$

Depending on the sign of C , we integrate accordingly. Suppose $C = 1$. Then:

$$\int \frac{dy}{y^2 + 1} = t + k \Rightarrow \tan^{-1}(y) = t + k \Rightarrow y = \tan(t + k).$$

This gives a general family of solutions to the original nonlinear ODE.

Chapter 3

Power Series

In this chapter we will explore a different approach to solving an ordinary differential equation. We have developed many methods in the previous chapters for obtaining the solution to an ordinary differential equation; in this chapter, we will present a method based upon the power series representation of a function. In previous chapters, we have not discussed (in general) a method of solution for an ordinary differential equation with variable coefficients; series methods provide one possible method of solution.

3.1 Introduction to Power Series

In the study of ODEs, power series methods provide a powerful tool for finding solutions, especially when standard techniques like separation of variables or integrating factors are not applicable. This approach involves expressing the solution as an infinite series of powers of the independent variable, typically centered at a point $x = x_0$. The general form of power series is written as:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

where a_n are constants (coefficients) to be determined and x_0 is the point about which the series is expanded (often called the ordinary point).

This method is particularly useful in solving linear differential equations with variable coefficients, which cannot be handled easily by elementary methods. If the point x_0 is an ordinary point of the differential equation (i.e., the coefficients of the equation are analytic at x_0), then a solution in the form of a power series

generally exists and can be found by substituting the series into the equation and solving for the coefficients a_n recursively.

3.1.1 Power Series Method

We begin with a discussion of the basic properties of a power series. A power series in $(x - a)$ is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x - a)^n = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots \quad (3.1)$$

where $c_0, c_1, \dots$, are constants, called the coefficients of the series; a is a constant, called the center of the series; and x is an independent variable[10]. In particular, the power series centered at zero ($a = 0$) has the form

$$\sum_{n=0}^{\infty} c_n(x)^n = c_0 + c_1(x) + c_2(x)^2 + \cdots \quad (3.2)$$

It is a very important first observation to note that all polynomials are (finite termed) power series.

A series of the form Equation 3.1.1 can always be reduced to the form equation 3.1.1 by substituting $X = x - a$. This substitution is merely a translation of the coordinate system. It is easy to see that the behavior of 3.1.1 near zero is exactly the same as the behavior of Equation 3.1.1 near a . For this reason we need only study the properties of series of the form 3.1.1

The fundamental assumption made in solving a linear differential equation of the form

$$f(x, y, y', y'', \dots) = 0$$

by the power series method is that *the solution of the differential equation can be expressed in the form of a power series*, say

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots \quad (9)$$

Since a power series can be differentiated, term by term, it follows that $y', y'', \dots$ can be obtained as

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + \cdots \quad (10)$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} = 2c_2 + 3 \cdot 2c_3 x + \dots \quad (11)$$

These series can be substituted into the given differential equation and the terms involving like powers of x are then collected. Such an expression would be of the form.

$$k_0 + k_1 x + k_2 x^2 + \dots = \sum_{n=0}^{\infty} k_n x^n = 0, \quad (12)$$

where $k_0, k_1, k_2, \dots$ are expressions involving the unknown coefficients $c_0, c_1, c_2, \dots$. Now Equation 3.1.1 must hold for all values of x in the interval of convergence; so all the coefficients $k_0, k_1, k_2, \dots$ must be zero, that is,

$$k_0 = 0 \quad k_1 = 0 \quad k_2 = 0 \quad \dots$$

Using these equations, we may be able to determine the coefficients $c_0, c_1, c_2, \dots$ of the power series solution y . We illustrate this method in the next several examples.

Examples 3.1.1 Solve the differential equation

$$(x^2 - 1)y'' + 3xy' + xy = 0$$

with $y(0) = 4$ and $y'(0) = 6$

Solution By using power series written in summation notation

$$y = \sum_{n=0}^{\infty} c_n x^n$$

This gives

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

And

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}$$

and then we are going to replace y'' , y' and y into equation

$$(x^2 - 1) \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + 3x \sum_{n=1}^{\infty} n c_n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0$$

Thus

$$\sum_{n=2}^{\infty} n(n-1)c_n x^n - \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+1} = 0,$$

so that reindexing the second and fourth series gives

$$\sum_{n=2}^{\infty} n(n-1)c_n x^n - \sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2} x^n + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=1}^{\infty} c_{n-1} x^n = 0.$$

Combining like powers of x and writing just a single series gives

$$-2c_2 + (c_0 + 3c_1 - 6c_3)x + \sum_{n=2}^{\infty} [-(n+2)(n+1)c_{n+2} + n(n+2)c_n + c_{n-1}] x^n = 0.$$

Comparing coefficients we have

$$-2c_2 = 0, \quad c_0 + 3c_1 - 6c_3 = 0$$

and

$$-(n+2)(n+1)c_{n+2} + n(n+2)c_n + c_{n-1} = 0 \quad \text{for } n \geq 2.$$

This gives $c_2 = 0$ and $c_3 = \frac{1}{6}c_0 + \frac{1}{2}c_1$, and the recurrence relation

$$c_{n+2} = \frac{n(n+2)c_n + c_{n-1}}{(n+1)(n+2)} \quad \text{for } n \geq 2.$$

Applying this recurrence relation successively yields

$$c_4 = \frac{8c_2 + c_1}{12} = \frac{1}{12}c_1, \quad c_5 = \frac{15c_3 + c_2}{20} = \frac{1}{8}c_0 + \frac{3}{8}c_1.$$

so that

$$y = c_0 + c_1 x + \left(\frac{c_0}{6} + \frac{c_1}{2}\right) x^3 + \frac{c_1}{12} x^4 + \left(\frac{c_0}{8} + \frac{3c_1}{8}\right) x^5 + \dots$$

This series can be written as

$$y = c_0 \left(1 + \frac{1}{6}x^3 + \frac{1}{8}x^5 + \cdots \right) + c_1 \left(x + \frac{1}{2}x^3 + \frac{1}{12}x^4 + \frac{3}{8}x^5 + \cdots \right)$$

Now applying the initial condition $y(0) = 4$ gives

$$c_0 = 4$$

and differentiating the series gives

$$y' = c_0 \left(\frac{1}{2}x^2 + \frac{5}{8}x^4 + \cdots \right) + c_1 \left(1 + \frac{3}{2}x^2 + \frac{1}{3}x^3 + \frac{15}{8}x^4 + \cdots \right)$$

so that $y'(0) = 6$ gives

$$c_1 = 6.$$

Thus, the power series solution that satisfies the initial-value problem is

$$\begin{aligned} y &= 4 \left(1 + \frac{1}{6}x^3 + \frac{1}{8}x^5 + \cdots \right) + 6 \left(x + \frac{1}{2}x^3 + \frac{1}{12}x^4 + \frac{3}{8}x^5 + \cdots \right) \\ &= 4 + 6x + \frac{11}{3}x^3 + \frac{1}{2}x^4 + \frac{11}{4}x^5 + \cdots \end{aligned}$$

3.1.2 Ordinary and Singular Points

We study second order linear homogeneous differential equations with variable coefficients,

$$y'' + p(x)y' + q(x)y = 0.$$

We look for solutions on a domain where the equation coefficients p, q are analytic functions. Recall that a function is analytic on a given domain iff it can be written as a convergent power series expansions on that domain. The power series method we have considered in the previous section sometimes fails to yield a solution for one equation while working very well for an apparently similar equation.

Definition 3.1.1 *The point $x = x_0$ is called an ordinary point of the differential equation when both $p(x)$ and $q(x)$ are analytic at $x = x_0$. If $x = x_0$ is not an ordinary point, it is called a singular point of the differential equation.*

Remark 3.1.1

Near a regular point x_0 the coefficients p and q in the differential equation above can be written in terms of power series centered at x_0 ,

$$p(x) = p_0 + p_1(x - x_0) + p_2(x - x_0)^2 + \cdots = \sum_{n=0}^{\infty} p_n(x - x_0)^n$$

,

$$q(x) = q_0 + q_1(x - x_0) + q_2(x - x_0)^2 + \cdots = \sum_{n=0}^{\infty} q_n(x - x_0)^n$$

and these power series converge in a neighborhood of x_0 .

Examples 3.1.2 *Solve the differential equation*

$$(1 - x^2)y'' - xy' + \mu^2 y = 0$$

Solution: Rewriting the equation as

$$y'' - \frac{x}{1 - x^2}y' + \frac{\mu^2}{1 - x^2}y = 0$$

gives $x = \pm 1$ as singular points. Thus, on the interval $(-1, 1)$ letting

$$\sum_{n=0}^{\infty} c_n x^n$$

and substituting into the differential equation gives

$$(1 - x^2) \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - x \sum_{n=1}^{\infty} c_n n x^{n-1} + \mu^2 \sum_{n=0}^{\infty} c_n x^n = 0.$$

Simplifying this expression and rewriting gives

$$\sum_{n=0}^{\infty} [c_{n+2}(n+2)(n+1) - c_n(n^2 - \mu^2)] x^n = 0.$$

Comparing coefficients gives the recurrence relation

$$c_{n+2} = \frac{(n^2 - \mu^2)}{(n+1)(n+2)} c_n, \quad \text{for } n = 0, 1, 2, \dots$$

This recurrence relation yields the even terms as

$$c_2 = \frac{-\mu^2 c_0}{2}, \quad c_4 = \frac{-(4 - \mu^2)\mu^2 c_0}{24}, \quad \dots$$

and the odd terms

$$c_3 = \frac{(1 - \mu^2)c_1}{6}, \quad c_5 = \frac{(1 - \mu^2)(9 - \mu^2)c_1}{120}, \quad \dots$$

so that the general solution is

$$y(x) = c_0 \left(1 + \frac{-\mu^2}{2}x^2 + \frac{-(4 - \mu^2)\mu^2}{24}x^4 + \dots \right) + c_1 \left(x + \frac{1 - \mu^2}{6}x^3 + \frac{(1 - \mu^2)(9 - \mu^2)}{120}x^5 + \dots \right).$$

If μ is an even integer, we see that all even coefficients having index greater than μ are zero. For instance, when $\mu = 4$ we have

$$c_6 = \frac{(4^2 - 4^2)}{30}c_4 = 0$$

so that $c_8 = 0, c_{10} = 0, \dots$. Thus when $\mu = 4$ the recurrence relation gives

$$c_2 = -8c_0, \quad c_4 = -c_2 = 8c_0, \quad c_6 = c_8 = c_{10} = \dots = 0.$$

Using these values, we have the fourth-degree polynomial

$$P(x) = c_0[1 - 8x^2 + 8x^4]$$

as a solution of Chebyshev equation

3.1.3 Solutions Near Regular Points

In the study of differential equations, a regular point refers to a point where the coefficients of the equation are analytic (i.e., expressible as a convergent power series), and the leading coefficient of the highest-order derivative does not vanish. Near such a point, solutions to linear differential equations can often be expressed as power series. This is a key idea behind methods like the Frobenius method, which assumes that if the point is regular, the solution can be written in the form of a power series with a positive radius of convergence.

Solving near regular points is crucial because it guarantees the existence and uniqueness of analytic solutions. It also allows for the application of standard techniques such as Taylor series expansion. These solutions are stable and well-behaved around the regular point, making it easier to approximate them numerically or analytically.

Definition 3.1.2 A point $x_0 \in \mathbb{R}$ is called a regular point of the equation

$$y'' + p(x)y' + q(x)y = 0$$

iff p, q are analytic function at x_0 . Otherwise x_0 is called a singular point of the equation

Examples 3.1.3 Consider the second-order linear differential equation:

$$(1 + x^2)y'' + 2xy' + y = 0.$$

Solution

First, Let rewrite the equation in standard form

$$y'' + p(x)y' + q(x)y = 0, \quad y'' + \frac{2x}{1+x^2}y' + \frac{1}{1+x^2}y = 0.$$

We examine whether $x_0 = 0$ is a regular point.

- The function

$$p(x) = \frac{2x}{1+x^2}$$

is analytic at $x = 0$, since it is a quotient of polynomials with a nonzero denominator at $x = 0$.

- Similarly,

$$q(x) = \frac{1}{1+x^2}$$

is analytic at $x = 0$.

Therefore, by Definition 3.1.1, since both $p(x)$ and $q(x)$ are analytic at $x = 0$, it follows that:

$x_0 = 0$ is a regular point of the differential equation.

Definition 3.1.3 Consider the differential equation

$$f_0(x)\frac{d^2y}{dx^2} + f_1(x)\frac{dy}{dx} + f_2(x)y = 0 \tag{3.1}$$

and assume that at least one of the functions

$$p(x) = \frac{f_1(x)}{f_0(x)}, \quad q(x) = \frac{f_2(x)}{f_0(x)}$$

is not analytic at x_0 , so that x_0 is a singular point of the differential equation (8.25).
If the functions defined by the products

$$(x - x_0)p(x) \quad \text{and} \quad (x - x_0)^2q(x)$$

are both analytic at x_0 , then x_0 is called a regular singular point of the differential equation (8.25). If either or both of the functions defined above are not analytic at x_0 , then x_0 is called an irregular singular point of (8.25).

Examples 3.1.4 the differential equation

$$x^2(x - 2)^2 \frac{d^2y}{dx^2} + 2(x - 2) \frac{dy}{dx} + (x + 1)y = 0$$

this equation can be written as

$$\frac{d^2y}{dx^2} + \frac{2dy}{x^2(x - 2)dx} + \frac{(x + 1)y}{x^2(x - 2)^2} = 0$$

so that

$$p(x) = \frac{2}{x^2(x - 2)} \quad q(x) = \frac{x + 1}{x^2(x - 2)}$$

are not analytic at $x = 0$ and $x = 2$. Thus $x = 0$ and $x = 2$ are singular points of the differential equation (8.28). Now at $x = 0$,

we consider the products

$$xp(x) = \frac{2}{x(x - 2)} \quad x^2q(x) = \frac{x + 1}{(x - 2)}$$

here $x^2q(x)$ is analytic at $x = 0$, while $xp(x)$ is not analytic at $x = 0$. Thus $x = 0$ is an irregular singular point of (8.28). Next considering the point $x = 2$, the products

$$(x - 2)p(x) = \frac{2}{x^2}, \quad (x - 2)^2q(x) = \frac{x + 1}{x^2}$$

are both analytic at $x = 2$ so that $x = 2$ is a regular singular point of (8.28).

The power series method provides a systematic approach for solving second-order ordinary differential equations, especially when standard analytical methods are not applicable. These ideas lead us to develop a method of obtaining a series solution when the differential equation has a regular singular point. We will develop this method in the next chapter.

Chapter 4

Application of Frobenius Method for solving Legendre Equation

Georg Frobenius was born in Charlottenburg, Germany in 1849. He received his doctorate at the University of Berlin (awarded with distinction) in 1870 under the supervision of Weierstrass. In 1874, after having taught at the secondary school level for several years, he was appointed to the University of Berlin as an extraordinary professor of mathematics. Frobenius made enormous contributions to group theory, particularly group representations, linear algebra, and analysis.

The most striking aspect of his mathematical practice is his extraordinary skill at calculations. In fact, Frobenius tried to solve mathematical problems to a large extent by means of a calculative, algebraic approach. Even his analytical work was guided by algebraic and linear algebraic methods. For Frobenius, conceptual argumentation played a somewhat secondary role. Although he argued in a comparatively abstract setting, abstraction was not an end in itself. Its advantages to him seemed to lie primarily in the fact that it can lead to much greater clearness and precision.

4.1 Introduction to Frobenius Method

In the study of differential equations, especially those with variable coefficients, standard power series methods often fail near singular points. To address this limitation, the Frobenius method provides a powerful extension that allows for the solution of linear second-order differential equations around regular singular points. This technique is particularly useful in mathematical physics and engineer-

ing, where such equations frequently arise. One notable application is in solving the Bessel equation, which models various physical phenomena such as heat conduction in cylindrical objects and wave propagation. In this chapter, we introduce the Frobenius method, laying the foundation for its application to the Legendre equation.

Theorem 4.1.1 *If x_0 is a regular singular point of the differential equation*

$$f_0(x)\frac{d^2y}{dx^2} + f_1(x)\frac{dy}{dx} + f_2(x)y = 0$$

, then this differential equation has at least one nontrivial solution of the form

$$|x - x_0|^r \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

, where r is a constant (that can be determined) and this solution is valid in some (deleted) interval $0 < |x - x_0| < R$, $R > 0$, about x_0 .

Examples 4.1.1 . Find the solution y near the regular singular point $x_0 = 0$ of the equation

$$x^2 \frac{d^2y}{dx^2} - x(x+3) \frac{dy}{dx} + (x+3)y = 0$$

in some interval $0 < x < R$

Solution: First let's rewrite the equation in the standard form

$$\frac{d^2y}{dx^2} - \frac{(x+3)}{x} \frac{dy}{dx} + \frac{(x+3)}{x^2} y = 0$$

since $x = 0$ is a regular point we know that

$$y = |x|^r \sum_{n=0}^{\infty} c_n (x)^n$$

, is a solution on $0 < |x| < R$

thus for $0 < x < R$ we seek a solution of the form

$$|x|^r \sum_{n=0}^{\infty} c_n (x)^n = \sum_{n=0}^{\infty} c_n x^{n+r}$$

now we need to find y', y''

$$\frac{dy}{dx} = \sum_{n=0}^{\infty} c_n(n+r)x^{n+r-1}$$

and

$$\frac{d^2y}{dx^2} = \sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r-2}$$

In the case $r = 0$ we had the relation $\sum_{n=0}^{\infty} n c_n x^{(n-1)} = \sum_{n=1}^{\infty} n c_n x^{(n-1)}$. But in our case $r \neq 0$, so we do not have the freedom to change in this way the starting value of the summation index n . If we want to change the initial value for n , we have to re-label the summation index. We now introduce these expressions into the differential equation. It is convenient to do this step by step. We start with the term $(x+3)y$, which has the form,

$$\begin{aligned} (x+3)y &= (x+3) \sum_{n=0}^{\infty} a_n x^{n+r} \\ &= \sum_{n=0}^{\infty} a_n x^{(n+r+1)} + \sum_{n=0}^{\infty} 3a_n x^{(n+r)} = \sum_{n=1}^{\infty} a_{n-1} x^{(n+r)} + \sum_{n=0}^{\infty} 3a_n x^{(n+r)}. \end{aligned}$$

We continue with the term containing y' ,

$$\begin{aligned} -x(x+3)y' &= -(x^2+3x) \sum_{n=0}^{\infty} (n+r)a_n x^{(n+r-1)} \\ &= -\sum_{n=0}^{\infty} (n+r)a_n x^{(n+r+1)} - \sum_{n=0}^{\infty} 3(n+r)a_n x^{(n+r)} \\ &= -\sum_{n=1}^{\infty} (n+r-1)a_{n-1} x^{(n+r)} - \sum_{n=0}^{\infty} 3(n+r)a_n x^{(n+r)}. \end{aligned}$$

Then, we compute the term containing y'' as follows,

$$\begin{aligned} x^2 y'' &= x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{(n+r-2)} \\ &= \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{(n+r)}. \end{aligned}$$

the guiding principle to rewrite each term is to have the power function $x^{(n+r)}$ labeled in the same way on every term this means that we do not have a sum involving terms with factors $x^{(n+r-1)}$ or factors $x^{(n+r+1)}$

Then, the differential equation can be written as follows,

$$\begin{aligned} & \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{(n+r)} - \sum_{n=1}^{\infty} (n+r-1)a_{n-1} x^{(n+r)} \\ & - \sum_{n=0}^{\infty} 3(n+r)a_n x^{(n+r)} + \sum_{n=1}^{\infty} a_{n-1} x^{(n+r)} + \sum_{n=0}^{\infty} 3a_n x^{(n+r)} = 0. \end{aligned}$$

In the equation above we need to split the sums containing terms with $n \geq 0$ into the term $n = 0$ and a sum containing the terms with $n \geq 1$, that is,

$$\begin{aligned} & = [r(r-1) - 3r + 3] a_0 x^r + \\ & \sum_{n=1}^{\infty} [(n+r)(n+r-1)a_n - (n+r-1)a_{n-1} - 3(n+r)a_n + a_{n-1} + 3a_n] x^{(n+r)} = 0, \end{aligned}$$

and this expression can be rewritten as follows,

$$\begin{aligned} & = [r(r-1) - 3r + 3] a_0 x^r + \\ & \sum_{n=1}^{\infty} [((n+r)(n+r-1) - 3(n+r) + 3) a_n - (n+r-1)a_{n-1}] x^{(n+r)} = 0 \end{aligned}$$

and then,

$$\begin{aligned} & [r(r-1) - 3r + 3] a_0 x^r + \\ & \sum_{n=1}^{\infty} [((n+r)(n+r-1) - 3(n+r) + 3) a_n - (n+r-2)a_{n-1}] x^{(n+r)} = 0 \end{aligned}$$

hence,

$$[r(r-1) - 3r + 3] a_0 x^r + \sum_{n=1}^{\infty} [(n+r-1)(n+r-3)a_n - (n+r-2)a_{n-1}] x^{(n+r)} = 0.$$

The indicial equation and the recurrence relation are given by the equations

$$r(r-1) - 3r + 3 = 0$$

and

$$(n+r-1)(n+r-3)a_n - (n+r-2)a_{(n-1)} = 0$$

The way to solve these equations in (3.2.6)–(3.2.7) is the following: First, solve Eq. (3.2.6) for the exponent r , which in this case has two solutions $r_{\pm}$; second, introduce the first solution r_+ into the recurrence relation in Eq. (3.2.7) and solve for the coefficients a_n ; the result is a solution y_+ of the original differential equation; then introduce the second solution r_- into Eq. (3.2.7) and solve again for the coefficients a_n ; the new result is a second solution y_- . Let us follow this procedure in the case of the equations above:

$$r^2 - 4r + 3 = 0 \quad \Rightarrow \quad r_{\pm} = \frac{1}{2} (4 \pm \sqrt{16 - 12}) \Rightarrow r_+ = 3, \quad r_- = 1.$$

Introducing the value $r_+ = 3$ into Eq. (3.2.7), we obtain:

$$(n+2)n a_n - (n+1)a_{n-1} = 0.$$

One can check that the solution y_+ obtained from this recurrence relation is given by:

$$y_+(x) = a_0 x^3 \left(1 + \frac{2}{3}x + \frac{1}{4}x^2 + \frac{1}{15}x^3 + \dots \right)$$

4.2 Overview and Standard Form of the Legendre Differential Equation

The Legendre equation appears when one solves the Laplace equation in spherical coordinates. The Laplace equation describes several phenomena, such as the static electric potential near a charged body, or the gravitational potential of a planet or star. When the Laplace equation describes a situation having spherical symmetry it makes sense to use spherical coordinates to solve the equation. It is in that case that the Legendre equation appears for a variable related to the polar angle in the spherical coordinate system[7].

The standard form of the Legendre differential equation is given by

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + l(l + 1)y = 0$$

where

$y = y(x)$ is the unknown function, and n is a non-negative constant, typically an integer. The variable x is usually defined on the interval $-1 < x < 1$, where the equation exhibits regular behavior. The points $x = \pm 1$ are regular singular points, which makes the equation suitable for techniques like the Frobenius method. When n is a non-negative integer, the solutions to this equation become polynomials of degree n , known as Legendre polynomials. These polynomials form an orthogonal set over the interval $[-1, 1]$ and are widely used in physics, particularly in potential theory, quantum mechanics, and numerical analysis such as Gaussian quadrature. Before applying the Frobenius method to this equation, it is essential to understand its structure, the nature of its singular points, and the types of solutions it admits depending on the value of n .

4.2.1 Application of Frobenius method for solving Legendre Equation

To apply the Frobenius method to the Legendre differential equation, we first consider the standard form of the equation :

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + l(l + 1)y = 0.$$

We start writing the equation in the standard form,

$$y'' - \frac{2x}{(1 - x^2)} y' + \frac{l(l + 1)}{(1 - x^2)} y = 0.$$

It is clear that the coefficient functions

$$p(x) = \frac{-2x}{(1 - x^2)} \quad q(x) = \frac{l(l + 1)}{(1 - x^2)}$$

are analytic on the interval $|x| < 1$, which is centered at $x_0 = 0$. Theorem 3.1.2 says that there are two solutions linearly independent and analytic on that interval. So we write the solution as a power series centered at $x_0 = 0$,

Let assume a solution of the form

$$y(x) = (x - x_0)^r \sum_{n=0}^{\infty} a_n (x - x_0)^n = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$$

and then for $r = 0$ we get

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

where $a_0 \neq 0$

and we compute its derivative,

$$y'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n,$$

where the first equality is the plain derivative, in the second we start the sum at $n = 1$ since the first term in the sum is zero, and in the third equality we rename the summation index $n \rightarrow n-1$, so when the old index starts at one, the new index starts at zero. The second derivative of y is treated in a similar way,

$$y''(x) = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

Then we continue working as follows,

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n,$$

$$-x^2 y'' = \sum_{n=0}^{\infty} -(n-1) n a_n x^n,$$

$$-2x y' = \sum_{n=0}^{\infty} -2n a_n x^n,$$

$$l(l+1)y = \sum_{n=0}^{\infty} l(l+1) a_n x^n.$$

The Legendre equation says that the addition of the four equations above must be zero,

$$\sum_{n=0}^{\infty} ((n+2)(n+1) a_{n+2} - (n-1) n a_n - 2n a_n + l(l+1) a_n) x^n = 0.$$

Therefore, every term in that sum must vanish,

$$(n+2)(n+1)a_{n+2} - (n-1)na_n - 2na_n + l(l+1)a_n = 0, \quad n \geq 0.$$

This is the recurrence relation for the coefficients a_n . After a few manipulations the recurrence relation becomes

$$a_{n+2} = \frac{(l-n)(l+n+1)}{(n+2)(n+1)}a_n, \quad n \geq 0.$$

By giving values to n we obtain,

$$a_2 = \frac{l(l+1)}{2!}a_0, \quad a_3 = \frac{(l-1)(l+2)}{3!}a_1.$$

Since a_4 is related to a_2 and a_5 is related to a_3 , we get,

$$a_4 = \frac{(l-2)(l+3)}{(3)(4)}a_2 = \frac{(l-2)(l+1)(l+3)}{4!}a_0,$$

$$a_5 = \frac{(l-3)(l+4)}{(4)(5)}a_3 = \frac{(l-3)(l-1)(l+2)(l+4)}{5!}a_1.$$

If one keeps solving the coefficients a_n in terms of either a_0 or a_1 , one gets the expression,

$$y(x) = a_0 \left[1 - \frac{l(l+1)}{2!}x^2 + \frac{(l-2)(l+1)(l+3)}{4!}x^4 + \dots \right] + a_1 \left[x - \frac{(l-1)(l+2)}{3!}x^3 + \frac{(l-3)(l-1)(l+2)(l+4)}{5!}x^5 + \dots \right].$$

Hence, the fundamental solutions are

$$y_1(x) = 1 - \frac{l(l+1)}{2!}x^2 + \frac{(l-2)(l+1)(l+3)}{4!}x^4 + \dots,$$

$$y_2(x) = x - \frac{(l-1)(l+2)}{3!}x^3 + \frac{(l-3)(l-1)(l+2)(l+4)}{5!}x^5 + \dots.$$

The ratio test provides the interval where the series above converge. For function y_1 we get, replacing n by $2n$,

$$\left| \frac{a_{2n+2}x^{2n+2}}{a_{2n}x^{2n}} \right| = \left| \frac{(l-2n)(l+2n+1)}{(2n+1)(2n+2)} \right| |x|^2 \rightarrow |x|^2 \quad \text{as } n \rightarrow \infty.$$

A similar result holds for y_2 . So both series converge on the interval defined by $|x| < 1$.

Remark: The functions y_1, y_2 are called *Legendre functions*. For a non-integer value of the constant l these functions cannot be written in terms of elementary functions. But when l is an integer, one of these series terminates and becomes a polynomial. The case l being a nonnegative integer is specially relevant in physics. For l even the function y_1 becomes a polynomial while y_2 remains an infinite series. For odd l the function y_2 becomes a polynomial while the y_1 remains an infinite series. For example, for $l = 0, 1, 2, 3$ we get,

$$\begin{aligned} l = 0, \quad y_1(x) &= 1, \\ l = 1, \quad y_2(x) &= x, \\ l = 2, \quad y_1(x) &= 1 - 3x^2, \\ l = 3, \quad y_2(x) &= x - \frac{5}{3}x^3. \end{aligned}$$

particular chosen inner product. We just say here that the first four polynomials are

$$\begin{aligned} l = 0, \quad y_1(x) &= 1, & P_0 &= y_1, & P_0(x) &= 1, \\ l = 1, \quad y_2(x) &= x, & P_1 &= y_2, & P_1(x) &= x, \\ l = 2, \quad y_1(x) &= 1 - 3x^2, & P_2 &= -\frac{1}{2}y_1, & P_2(x) &= \frac{1}{2}(3x^2 - 1), \\ l = 3, \quad y_2(x) &= x - \frac{5}{3}x^3, & P_3 &= -\frac{5}{2}y_2, & P_3(x) &= \frac{7}{2}(5x^3 - 3x). \end{aligned}$$

These polynomials, P_n , are called *Legendre polynomials*.

We consider the Legendre differential equation:

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0,$$

where $n \in \mathbb{N}_0$. We seek a solution of the form:

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

Differentiating term by term:

$$y'(x) = \sum_{k=0}^{\infty} a_k (k + r) x^{k+r-1}, \quad y''(x) = \sum_{k=0}^{\infty} a_k (k + r)(k + r - 1) x^{k+r-2}.$$

Substituting into the equation:

$$(1 - x^2) \sum_{k=0}^{\infty} a_k (k + r)(k + r - 1) x^{k+r-2} - 2x \sum_{k=0}^{\infty} a_k (k + r) x^{k+r-1} + n(n + 1) \sum_{k=0}^{\infty} a_k x^{k+r} = 0.$$

Expanding and grouping terms:

$$\begin{aligned} & \sum_{k=0}^{\infty} a_k(k+r)(k+r-1)x^{k+r-2} - \sum_{k=0}^{\infty} a_k(k+r)(k+r-1)x^{k+r} \\ & - 2 \sum_{k=0}^{\infty} a_k(k+r)x^{k+r} + n(n+1) \sum_{k=0}^{\infty} a_k x^{k+r} = 0. \end{aligned}$$

This simplifies to:

$$\sum_{k=0}^{\infty} a_k(k+r)(k+r-1)x^{k+r-2} + \sum_{k=0}^{\infty} [-a_k(k+r)(k+r-1) - 2a_k(k+r) + n(n+1)a_k] x^{k+r} = 0.$$

From the coefficient of the lowest-order term x^{r-2} , we obtain the indicial equation:

$$a_0 r(r-1) = 0 \quad \Rightarrow \quad r = 0 \text{ or } r = 1.$$

We choose $r = 0$, and obtain the recurrence relation from the coefficient of x^{k+r} (i.e., x^k):

$$a_k [k(k-1) + 2k - n(n+1)] + a_{k-2}(k-2)(k-3) = 0.$$

That is,

$$a_k = \frac{(k-2)(k-3)}{k(k-1) + 2k - n(n+1)} a_{k-2}.$$

This recurrence determines all coefficients a_k in terms of a_0 and a_1 . If n is a non-negative integer, the solution terminates (i.e., becomes a polynomial), and defines the Legendre polynomial $P_n(x)$.

4.2.2 Graphical and Numerical Illustration of Legendre Polynomials

To enhance understanding of the Legendre polynomials derived from solving Legendre's differential equation, it is helpful to visualize them. Legendre polynomials $P_n(x)$ are a family of orthogonal polynomials defined on the interval $[-1, 1]$, and each has distinct characteristics based on the degree n .

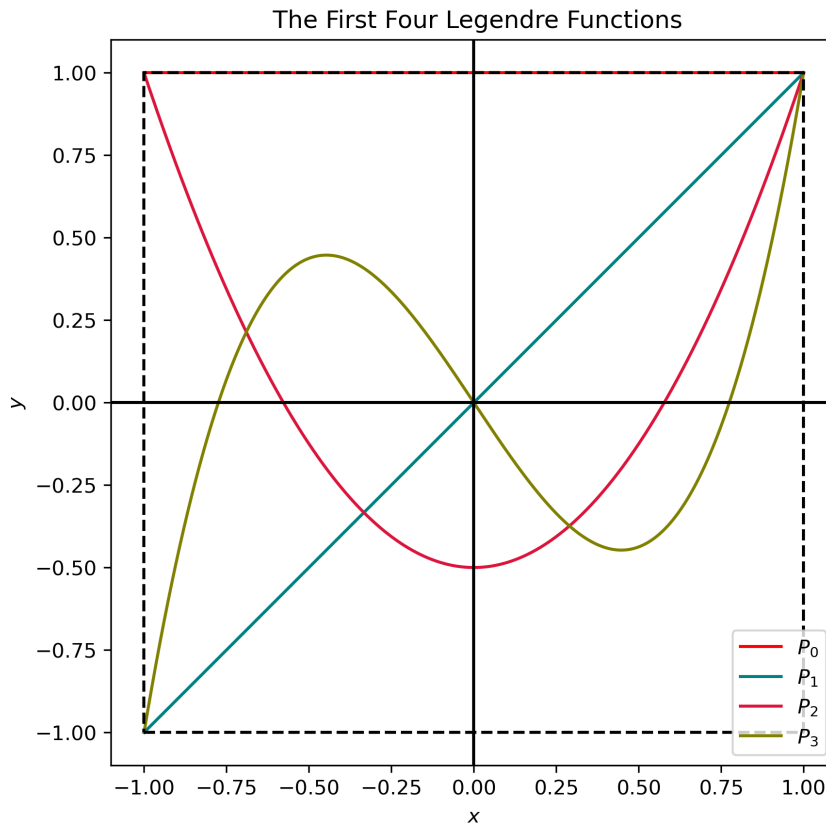
By plotting the first few polynomials—such as $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$, and $P_3(x) = \frac{1}{2}(5x^3 - 3x)$ —we can observe how their shapes evolve with increasing order. These plots not only demonstrate the symmetry and oscillatory behavior of the polynomials but also highlight their zeros, which are important in applications such as numerical integration (Gaussian quadrature).

4.2.3 Visualization of Legendre Polynomials Up to Degree 3

we use Numerical tool like Python's `matplotlib` library to generate these graphs, offering intuitive insight into the nature of these mathematical solutions.



Figure 4.1: A combined figure showing python code used to plot First four Legendre Polynomials.



This plot effectively visualizes the first four Legendre polynomials $P_0(x)$ to $P_3(x)$, which are solutions to Legendre's differential equation on the interval $[-1, 1]$. Each curve shows how the polynomials increase in complexity and oscillatory behavior with higher degrees. The symmetry and location of their zeros are clearly visible especially important in applications like Gaussian quadrature, where these zeros serve as integration nodes. The use of distinct colors and a well defined legend enhances clarity and supports a deeper understanding of the mathematical structure of these orthogonal functions.

Chapter 5

Conclusion and Recommendation

This part deals with the conclusion of this work, and the recommendations for the Department of Mathematics, and also for others who will do research in the same area.

5.1 Conclusion

This research has thoroughly explored the Frobenius method as a powerful tool for solving second-order linear ordinary differential equations with variable coefficients, especially near regular singular points. Through both theoretical explanation and practical application, particularly to the Legendre differential equation, the study has demonstrated that the Frobenius method effectively produces series solutions where traditional methods may fail or be inefficient.

The method's strength lies in its ability to handle singularities and provide analytical solutions in the form of power series, offering deep insights into the behavior of physical systems modeled by such equations. Visualization of the Legendre polynomials and their properties such as orthogonality and symmetry further illustrated the method's practical significance, especially in physics and engineering contexts like quantum mechanics, wave propagation, and numerical integration.

5.2 Recommendations

- **Further Application to Other Special Functions:** Researchers should apply the Frobenius method to other classes of differential equations such as

Bessel's and Hermite's equations to deepen understanding of special functions in applied mathematics.

- **Integration with Computational Tools:** Combining analytical methods like Frobenius with numerical computation tools such as MATLAB, Python (SymPy), or Mathematica can enhance solution accuracy and allow exploration of higher-order terms or complex systems.
- **Educational Inclusion:** The Frobenius method should be more deeply integrated into university-level applied mathematics curricula due to its wide applicability in mathematical physics and engineering.
- **Exploration of Nonlinear Extensions:** Future work can focus on adapting similar power series approaches to specific classes of nonlinear differential equations, potentially opening up new avenues in mathematical modeling.
- **Real-world Case Studies:** Apply the Frobenius method to real-world physical or engineering problems (e.g., electrical circuits, mechanical vibrations, fluid flow) to evaluate its practical efficiency and limitations beyond idealized examples.

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